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HOTEL BOOKING ANALYTICS

Booking, Cancellation, Customer Behaviour & Revenue Analysis

Dataset Source: Hotel Booking Demand — Kaggle
Tools Used: Python | SQL | Power BI

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End-to-End Data Analytics Portfolio Project

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1. Executive Summary

This project presents an end-to-end analysis of hotel booking data to examine booking patterns, cancellations, customer behaviour, and revenue performance across City Hotel and Resort Hotel. The objective was to transform raw booking data into meaningful business insights using Python, SQL, and Power BI.

Python was used for data inspection, cleaning, transformation, and exploratory data analysis to prepare a reliable dataset for further analysis. SQL was then used to query the cleaned data and investigate key business areas, including booking and cancellation behaviour, customer segments, revenue performance, and seasonal trends. Finally, Power BI was used to develop an interactive three-page dashboard covering Executive Overview, Booking & Cancellation Analysis, and Revenue & Seasonal Trends.

The analysis identified several important patterns in hotel performance. Across 119,209 bookings, the overall cancellation rate was 37.08%, representing 44,199 cancelled bookings. Completed stays generated approximately $25.99 million in estimated revenue, while the overall Average Daily Rate (ADR) was $101.93. The analysis also highlighted differences between hotel types, customer groups, lead-time categories, market segments, distribution channels, and monthly performance.

Overall, the project demonstrates how an integrated analytics workflow can convert hotel booking data into clear, actionable insights that can support revenue management, cancellation monitoring, customer analysis, and data-driven decision-making.

2. Project Overview

The Hotel Booking Analytics project is an end-to-end data analytics portfolio project designed to analyse hotel booking behaviour and transform raw booking data into meaningful business insights. The project focuses on two hotel types—City Hotel and Resort Hotel—and examines key areas including booking volume, cancellations, customer behaviour, revenue performance, Average Daily Rate (ADR), market segments, distribution channels, and seasonal booking patterns.

The project follows a complete analytics workflow using Python, SQL, and Power BI. Python was used to inspect, clean, transform, and prepare the raw dataset for analysis. The cleaned dataset was then analysed using SQL to answer business-focused questions and identify patterns across different booking and customer characteristics. Power BI was used as the final visualization layer to convert the analytical results into an interactive dashboard that allows hotel performance to be explored through key performance indicators (KPIs), charts, and filters.

The Power BI dashboard was organised into three analytical pages:

- **Executive Overview** – provides a high-level view of total bookings, cancellation performance, completed-stay revenue, ADR, and comparisons between City Hotel and Resort Hotel.
- **Booking & Cancellation Analysis** – examines cancellation behaviour across customer type, lead time, room-change status, repeat status, and month.
- **Revenue & Seasonal Trends** – analyses completed-stay revenue across hotels, months, market segments, and distribution channels, together with ADR performance.

By combining data preparation, querying, analysis, and visualization within a single workflow, the project demonstrates how hotel booking data can be used to understand operational and customer patterns and support more informed business decision-making.

3. Business Problem & Objectives

3.1 Business Problem

Hotels generate large volumes of booking data, but raw booking records alone provide limited value unless they are analysed to identify meaningful patterns. Hotel managers need to understand not only how many bookings are received, but also how frequently bookings are cancelled, which customer and booking characteristics are associated with cancellations, how revenue varies across different channels and market segments, and how performance changes throughout the year.

High cancellation rates can create challenges for occupancy planning and revenue management, while differences in customer behaviour, booking lead time, distribution channels, and seasonal demand can influence overall hotel performance. Understanding these patterns can help management identify areas that require closer attention and make more informed operational and commercial decisions. This project therefore focuses on analysing historical booking data from City Hotel and Resort Hotel to provide a clearer view of booking, cancellation, customer, and revenue performance.

3.2 Project Objectives

The main objective of the project is to transform hotel booking data into useful business insights through an end-to-end analytics workflow. Specifically, the project aims to:

- Analyse overall booking volume and compare performance between City Hotel and Resort Hotel.
- Measure cancellation rates and investigate how cancellations vary across hotel type, customer type, booking lead time, repeat status, room-change status, and month.
- Analyse completed-stay revenue and compare revenue performance between the two hotel types.
- Examine revenue contribution across market segments and distribution channels.
- Analyse monthly revenue patterns to identify seasonal changes in hotel performance.
- Compare Average Daily Rate (ADR) between City Hotel and Resort Hotel.
- Develop an interactive Power BI dashboard that enables users to explore key booking, cancellation, customer, and revenue metrics.
- Present the results in a clear format that can support data-driven hotel management and business decision-making.

Together, these objectives provide a structured framework for analysing the dataset and connecting the technical analysis with practical business questions.

4. Dataset Overview

4.1 Data Source

The dataset used for this project is based on the Hotel Booking Demand dataset originally published by Nuno António, Ana de Almeida, and Luís Nunes in 2019. For this portfolio project, the dataset was accessed through Kaggle.

The original dataset contains booking information for two types of hotels: a City Hotel and a Resort Hotel. Each record represents an individual hotel booking and contains information relating to the booking, customer characteristics, stay details, cancellation status, distribution channel, market segment, room allocation, and pricing.

4.2 Dataset Size and Time Period

The original Hotel Booking Demand data contains a total of 119,390 booking records:

- **City Hotel:** 79,330 bookings
- **Resort Hotel:** 40,060 bookings

The bookings cover scheduled arrival dates from July 2015 to August 2017 and include both completed and cancelled bookings. During the data preparation stage of this project, the raw dataset was inspected and cleaned in Python before being used for SQL analysis and Power BI visualization. Therefore, the final analytical dataset reflects the cleaning and transformation decisions applied during the project.

4.3 Key Data Areas

The dataset contains variables covering several important areas of hotel operations and customer booking behaviour, including:

- **Hotel information** – City Hotel or Resort Hotel
- **Booking status** – whether a booking was completed or cancelled
- **Arrival information** – year, month, week, and day of arrival
- **Lead time** – number of days between booking and scheduled arrival
- **Stay information** – weekday and weekend nights
- **Guest information** – adults, children, babies, and repeat-guest status
- **Customer type** – such as Transient, Contract, Transient-Party, and Group
- **Market segment and distribution channel**
- **Room information** – reserved and assigned room types
- **Average Daily Rate (ADR)** – average room revenue per occupied room for the booking
- **Booking changes and previous booking history**

These variables provide the foundation for analysing booking behaviour, cancellation patterns, customer characteristics, revenue performance, and seasonal trends.

4.4 Analytical Focus

For this project, the dataset was primarily used to investigate four areas:

Booking Performance: Understanding overall booking volumes and differences between City Hotel and Resort Hotel.

Cancellation Behaviour: Examining cancellation rates across hotel type, customer type, booking lead time, repeat status, room-change status, and month.

Customer Behaviour: Exploring how different customer and booking characteristics relate to hotel booking and cancellation patterns.

Revenue & Seasonal Performance: Analysing completed-stay revenue, ADR, market segments, distribution channels, hotel type, and monthly revenue trends.

These analytical areas were subsequently explored using Python and SQL and presented through the final Power BI dashboard.

5. Data Cleaning & Preparation - Python

Python was used as the primary tool for inspecting, cleaning, transforming, and preparing the Hotel Booking Demand dataset before further analysis in SQL and Power BI. The objective of this stage was to improve data quality, investigate unusual values, and create additional analytical variables required for the later stages of the project.

5.1 Initial Data Inspection

The dataset was first explored using Pandas to understand its structure, dimensions, data types, descriptive statistics, missing values, and overall data quality. Key numerical and categorical variables were reviewed to identify potential issues that could affect the analysis. Particular attention was given to booking status, guest information, Average Daily Rate (ADR), length of stay, room allocation, and other variables relevant to cancellation and revenue analysis.

5.2 Data Quality Assessment and Cleaning

The data quality assessment included checks for missing values, duplicate or repeated records, unusual numerical values, and potentially invalid booking records. One particularly unusual ADR value of 5,400 was identified during the investigation. Because this value could not be independently verified and was substantially higher than the remaining valid ADR values, it was treated as missing rather than replaced with an assumed value. As a result, one missing value was intentionally retained in the adr column.

The corresponding estimated_revenue value was also left missing because estimated revenue was calculated using ADR. This approach avoided introducing an artificial pricing or revenue value into the analysis. Additional checks were performed on guest counts, stay duration, zero-night bookings, and other potentially unusual observations before the dataset was prepared for analysis.

5.3 Data Transformation & Analysis Preparation

Several new variables were created to make the dataset more useful for business analysis.

Total Nights was calculated by combining weekday and weekend stays, providing a single measure of booking duration.

Total Guests was created by combining the number of adults, children, and babies associated with each booking.

Arrival Month Number was created to support correct chronological sorting of monthly results.

Estimated Revenue was calculated using ADR and total nights:

Estimated Revenue = ADR × Total Nights

A separate Completed-Stay Revenue variable was then created so that estimated revenue was counted only when the reservation status was recorded as Check-Out. Cancelled and other non-completed bookings were assigned zero completed-stay revenue.

Bookings were also grouped into five Lead Time Categories:

- 0–7 days
- 8–30 days
- 31–90 days
- 91–180 days
- 181+ days

A Room Type Changed indicator was created by comparing the reserved room type with the assigned room type. This allowed the analysis to compare cancellation behaviour between bookings where the assigned room remained the same and bookings where it changed. Additional analytical indicators were created for areas such as zero-night stays and parking requirements.

5.4 Final Data Validation

Following data cleaning and transformation, the prepared dataset was validated again to confirm its dimensions, data types, missing values, and newly created analytical fields. The final dataset used for the later stages of the project contained 119,209 booking records and 42 columns. The remaining missing ADR value and its corresponding estimated revenue value were intentionally retained rather than replaced with potentially inaccurate assumptions. The cleaned and transformed dataset was then exported for further analysis in SQL and subsequently used to support the Power BI dashboard.

6. Exploratory Data Analysis – Python

Following data cleaning and preparation, Exploratory Data Analysis (EDA) was conducted in Python to understand the main booking, cancellation, customer, pricing, and revenue patterns within the dataset. Pandas was used for data aggregation and analysis, while Matplotlib and Seaborn were used to visualize important relationships and trends.

6.1 Booking & Hotel Performance

Booking volumes were analysed across City Hotel and Resort Hotel to understand the distribution of demand between the two properties. Monthly booking patterns were also examined to identify changes in booking activity throughout the year. The analysis showed that booking activity varied

across months, indicating seasonal changes in hotel demand. City Hotel accounted for a larger share of total bookings, reflecting its greater representation within the dataset.

6.2 Cancellation Analysis

Cancellation behaviour was examined across several booking characteristics to identify patterns associated with higher or lower cancellation rates. City Hotel recorded a higher cancellation rate than Resort Hotel. Further analysis showed a particularly strong relationship between booking lead time and cancellation behaviour: bookings made further in advance had progressively higher cancellation rates. Cancellation rates were also compared across customer types. Transient customers recorded the highest cancellation rate at approximately 40.79%, followed by Contract customers at 30.99% and Transient-Party customers at 25.45%. Group customers recorded the lowest cancellation rate at approximately 10.10%. Room allocation was also examined by comparing the reserved room type with the room type ultimately assigned to the booking. Bookings with no room-type change recorded a substantially higher cancellation rate than bookings where the assigned room type differed from the reserved room type.

6.3 Booking Lead Time

Booking lead time was divided into five categories to investigate how far in advance customers booked and how this related to cancellation behaviour.

The cancellation rate increased consistently across the lead-time categories:

- **0–7 days:** 9.60%
- **8–30 days:** 27.89%
- **31–90 days:** 37.73%
- **91–180 days:** 44.74%
- **181+ days:** 57.01%

This pattern indicates that bookings made further in advance were associated with a greater likelihood of cancellation within this dataset.

6.4 Customer & Booking Behaviour

Customer-related variables were analysed to understand differences in booking behaviour across customer groups. The analysis considered customer type, repeat-guest status, guest composition, booking changes, special requests, and other booking characteristics. These comparisons provided additional context for understanding which types of bookings demonstrated different cancellation patterns and helped identify customer characteristics that could be investigated further for retention and cancellation management.

6.5 Pricing & Revenue Analysis

ADR and estimated completed-stay revenue were analysed to compare financial performance across hotel types and time periods. City Hotel generated approximately $14.39 million in estimated completed-stay revenue, compared with approximately $11.60 million for Resort Hotel. However, Resort Hotel recorded a higher average completed-stay revenue per booking within the analysis. Monthly ADR analysis also showed different pricing patterns between the two properties. Resort Hotel

displayed stronger seasonal pricing variation, particularly during July and August, while City Hotel pricing remained comparatively more consistent throughout the year.

6.6 EDA Outcome

The Python EDA identified several patterns that were subsequently investigated further using SQL and presented through the Power BI dashboard. In particular, cancellation behaviour, booking lead time, customer characteristics, hotel type, seasonality, ADR, and revenue emerged as important areas for deeper business analysis. The EDA therefore provided the analytical foundation for developing the SQL business queries and determining the key metrics and visualizations included in the final dashboard.

7. SQL Analysis

Following data cleaning, preparation, and exploratory analysis in Python, the cleaned hotel booking dataset was imported into SQL for structured business analysis. SQL was used to validate the prepared data, calculate key performance metrics, compare customer and booking segments, and answer business-focused questions identified during the Python analysis. The SQL analysis focused on booking and cancellation behaviour, seasonal patterns, customer characteristics, pricing, and revenue performance.

7.1 Data Validation

Before conducting the main analysis, SQL queries were used to validate the imported dataset and confirm that the record count and important analytical fields were consistent with the dataset prepared in Python. The SQL dataset contained 119,209 booking records, confirming that the cleaned data had been transferred successfully for further analysis.

7.2 Overall Booking & Cancellation Performance

SQL was used to calculate overall booking and cancellation performance. Of the 119,209 total bookings, 44,199 were cancelled, resulting in an overall cancellation rate of approximately 37.08%. Cancellation performance was then compared between the two hotel types. City Hotel recorded a cancellation rate of approximately 41.79%, while Resort Hotel recorded approximately 27.77%. This demonstrated a clear difference in cancellation behaviour between the two properties.

7.3 Lead Time & Cancellation Behaviour

The relationship between booking lead time and cancellation was examined using the lead-time categories created during Python data preparation.

The analysis showed a consistent increase in cancellation rates as booking lead time increased:

- **0–7 days:** 9.60%
- **8–30 days:** 27.89%
- **31–90 days:** 37.73%
- **91–180 days:** 44.74%
- **181+ days:** 57.01%

Bookings made more than 181 days in advance therefore showed the highest cancellation rate, while bookings made within seven days of arrival showed the lowest.

7.4 Customer Behaviour

SQL was also used to compare cancellation behaviour across different customer characteristics. Repeat-guest analysis showed that repeated guests had a cancellation rate of approximately 14.65%, compared with approximately 37.81% for non-repeated guests. Cancellation rates also varied across customer types. Transient customers recorded the highest cancellation rate, while Group customers recorded the lowest. These results highlighted meaningful differences in booking behaviour across customer segments.

7.5 Revenue & Hotel Performance

Revenue analysis focused on completed-stay revenue, ensuring that revenue comparisons reflected bookings that resulted in completed stays rather than cancelled reservations. City Hotel generated approximately $14.39 million in estimated completed-stay revenue, while Resort Hotel generated approximately $11.60 million. Combined, completed stays generated approximately $25.99 million in estimated revenue. Although City Hotel generated greater total completed-stay revenue, Resort Hotel recorded a higher average completed-stay revenue per booking, indicating differences in the revenue profile of the two properties.

7.6 Market Segment & Distribution Channel Analysis

SQL was used to analyse how completed-stay revenue was distributed across market segments and distribution channels. The analysis showed that the TA/TO (Travel Agent/Tour Operator) distribution channel accounted for approximately 80.05% of estimated completed-stay revenue, making it the dominant revenue-generating channel within the dataset. Revenue was also compared across market segments to understand which booking sources contributed most strongly to completed-stay revenue and to provide additional context for the Power BI revenue analysis.

7.7 Advanced SQL Analysis

In addition to summary-level analysis, more advanced SQL techniques were used to explore the dataset in greater depth. These included grouped aggregations, conditional calculations, ranking, and comparative analysis across hotels, customer groups, time periods, and revenue categories. These queries helped convert the cleaned dataset into business-focused results and provided an additional layer of validation for patterns initially identified during Python EDA.

7.8 SQL Analysis Outcome

The SQL analysis confirmed several important patterns identified during Python EDA while allowing the results to be investigated through structured business queries. In particular, hotel type, booking lead time, customer characteristics, market segments, distribution channels, and seasonality showed meaningful differences in cancellation and revenue performance. The SQL findings were subsequently used alongside the Python analysis to determine the KPIs, comparisons, and visualizations presented in the final Power BI dashboard.

8. Power BI Dashboard

Power BI was used as the final visualization and reporting layer of the project. The cleaned and analysed hotel booking data was transformed into an interactive three-page dashboard designed to present the most important booking, cancellation, customer, revenue, and seasonal insights in a clear and accessible format. The dashboard was structured into three analytical pages: Executive Overview,

Booking & Cancellation Analysis, and Revenue & Seasonal Trends. Interactive Hotel, Year, and Month slicers were included to allow users to explore performance across different hotels and time periods.

8.1 Executive Overview

The Executive Overview provides a high-level summary of overall hotel performance and allows the main KPIs to be understood at a glance.

The page presents five key performance indicators:

- **Total Bookings:** 119,209
- **Total Cancellations:** 44,199
- **Cancellation Rate:** 37.08%
- **Completed-Stay Revenue:** approximately $25.99 million
- **Average Daily Rate (ADR):** $101.93

In addition to the headline KPIs, the page compares total bookings, cancellation rates, and completed-stay revenue between City Hotel and Resort Hotel. City Hotel accounts for a larger number of bookings and generates higher total completed-stay revenue than Resort Hotel. However, City Hotel also records a higher cancellation rate, highlighting an important difference in booking performance between the two hotel types. The Executive Overview therefore provides a concise starting point for understanding overall hotel performance before exploring cancellation behaviour and revenue trends in greater detail on the subsequent dashboard pages.

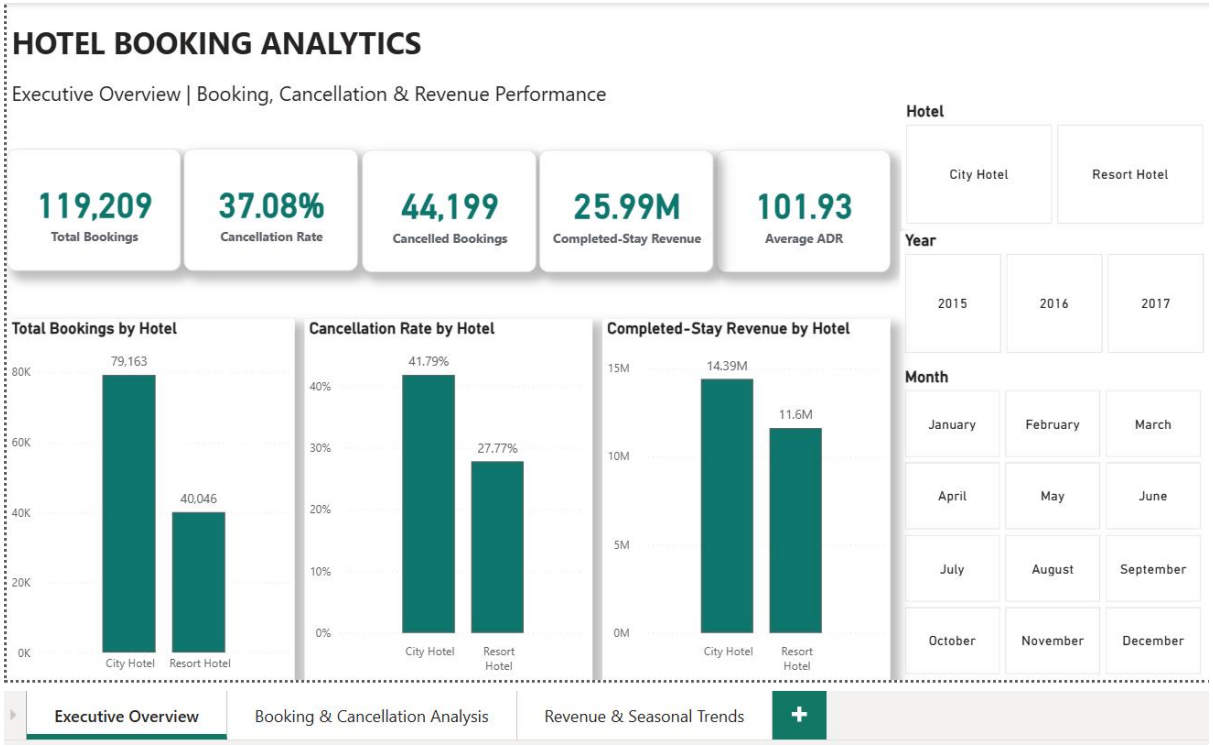


Figure 1: Executive Overview Dashboard

8.2 Booking & Cancellation Analysis

The Booking & Cancellation Analysis page provides a more detailed view of cancellation behaviour and examines how cancellation rates vary across different customer and booking characteristics.

The page analyses cancellation performance across:

- **Hotel Type**
- **Customer Type**
- **Booking Lead Time**
- **Month**
- **Repeat-Guest Status**
- **Room-Change Status**

One of the clearest patterns is the relationship between booking lead time and cancellations. The cancellation rate increases as bookings are made further in advance, rising from approximately 9.60% for bookings made 0–7 days before arrival to approximately 57.01% for bookings made 181 or more days in advance. Cancellation behaviour also varies across customer groups. Transient customers record the highest cancellation rate among the customer types, while repeated guests have a substantially lower cancellation rate than non-repeated guests. The room-change analysis provides an additional perspective by comparing bookings where the assigned room remained the same as the reserved room with bookings where the room was changed. Monthly analysis further helps identify how cancellation rates vary throughout the year. Overall, this dashboard page provides a focused view of the booking and customer characteristics associated with different cancellation patterns and helps identify areas that may require greater attention in cancellation management.

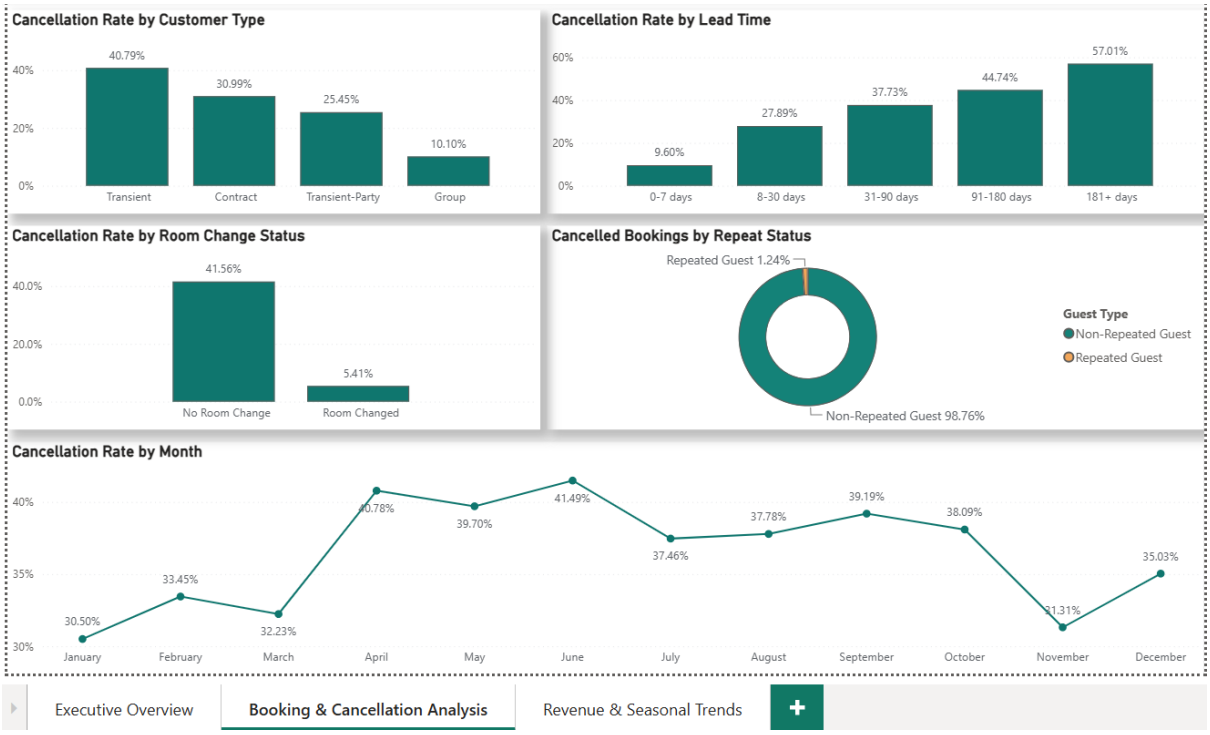


Figure 2: Booking & Cancellation Analysis Dashboard

8.3 Revenue & Seasonal Trends

The Revenue & Seasonal Trends page focuses on the financial performance of the hotels and examines how completed-stay revenue and Average Daily Rate (ADR) vary across different business dimensions.

The page analyses:

- **Completed-Stay Revenue by Month**
- **Completed-Stay Revenue by Hotel**
- **Average Daily Rate (ADR) by Hotel**
- **Completed-Stay Revenue by Market Segment**
- **Completed-Stay Revenue by Distribution Channel**

The analysis shows that City Hotel generates higher total completed-stay revenue than Resort Hotel, contributing approximately $14.39 million, compared with approximately $11.60 million for Resort Hotel. However, City Hotel records a higher overall Average Daily Rate (ADR) than Resort Hotel, while the two hotels show different seasonal pricing patterns. Monthly completed-stay revenue also varies throughout the year, providing insight into seasonal changes in hotel performance. Revenue comparisons across market segments and distribution channels further show how different booking sources contribute to overall financial performance. In particular, the TA/TO (Travel Agent/Tour Operator) distribution channel represents the largest share of completed-stay revenue, highlighting the importance of this channel within the analysed booking data. Overall, this page provides a financial and seasonal perspective on hotel performance and complements the booking and cancellation analysis presented on the previous dashboard page.

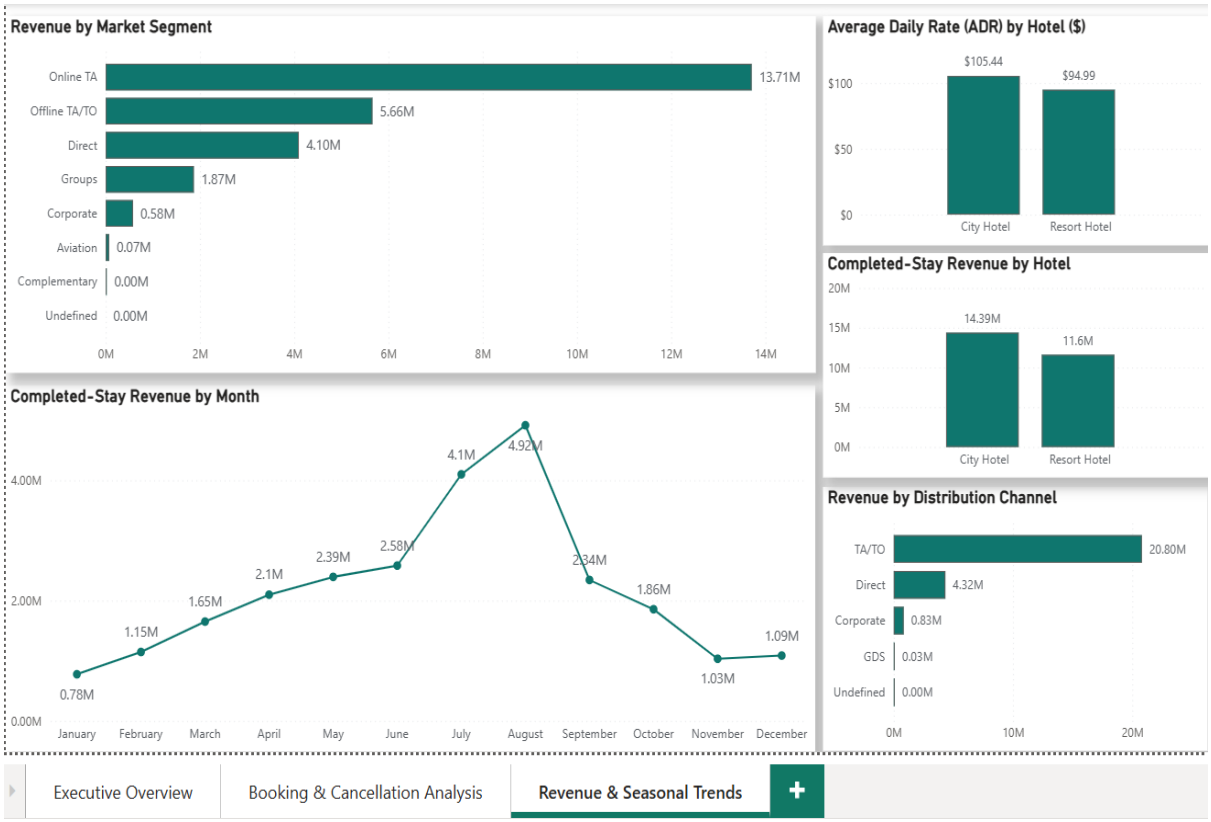


Figure 3: Revenue & Seasonal Trends Dashboard

8.4 Interactive Filtering

Interactive slicers were incorporated into the Power BI dashboard to allow users to explore hotel performance dynamically rather than viewing only the overall results.

The dashboard includes three main filters:

- **Hotel** – allows users to analyse City Hotel, Resort Hotel, or both.
- **Year** – allows performance to be examined across different years within the dataset.
- **Month** – enables users to focus on individual months or compare selected periods.

The Hotel and Year slicers are synchronized across all three dashboard pages, while the Month slicer is available on the Executive Overview page. These slicers allow the dashboard visuals and KPIs to respond to the selected criteria, making it possible to investigate booking, cancellation, and revenue performance for specific hotels and time periods. The interactive filtering functionality improves the usability of the dashboard by allowing users to move from a high-level overview to more focused analysis without changing the underlying report.

8.5 Dashboard Outcome

The Power BI dashboard brings together the findings generated through Python and SQL into a single interactive reporting environment. Its three-page structure separates overall performance, booking and cancellation behaviour, and revenue and seasonal trends, making the analysis easier to navigate and interpret. The dashboard combines key performance indicators (KPIs), comparative charts, trend analysis, and interactive filters to provide both a high-level overview and more detailed exploration of hotel performance. By transforming the analytical results into clear and interactive visualizations, the dashboard provides an accessible way to monitor booking activity, identify cancellation patterns, examine customer behaviour, and evaluate revenue performance across different hotels and time periods. Overall, the Power BI dashboard serves as the final reporting layer of the project and demonstrates how data prepared in Python and analysed through SQL can be converted into meaningful business insights for data-driven decision-making.

9. Key Findings & Business Insights

The combined analysis conducted in Python, SQL, and Power BI identified several important patterns across hotel bookings, cancellations, customer behaviour, and revenue performance. The following findings summarize the most significant results of the project and their potential business implications.

9.1 City Hotel Has a Higher Cancellation Rate

The analysis identified a clear difference in cancellation behaviour between the two hotel types. City Hotel recorded a cancellation rate of approximately 41.79%, compared with 27.77% for Resort Hotel. This indicates that although City Hotel receives a larger volume of bookings, a greater proportion of those bookings do not result in completed stays. From a business perspective, the higher cancellation rate may make occupancy and revenue planning more challenging for City Hotel. Management could therefore benefit from monitoring the characteristics of City Hotel bookings more closely, particularly factors such as booking lead time, customer type, distribution channel, and cancellation policies. This finding does not establish that hotel type itself causes cancellations, but it identifies City Hotel as an important area for further cancellation monitoring and analysis.

9.2 Longer Booking Lead Times Are Associated with Higher Cancellation Rates

One of the strongest patterns identified in the analysis was the relationship between booking lead time and cancellation rate. As the time between booking and scheduled arrival increased, the proportion of cancelled bookings also increased.

The cancellation rates across the five lead-time categories were:

- **0–7 days:** 9.60%
- **8–30 days:** 27.89%
- **31–90 days:** 37.73%
- **91–180 days:** 44.74%
- **181+ days:** 57.01%

Bookings made 181 or more days in advance recorded the highest cancellation rate, while bookings made within seven days of arrival recorded the lowest. From a business perspective, long-lead-time reservations may require closer monitoring because they are associated with greater cancellation uncertainty. Hotels could consider using booking reminders, reconfirmation strategies, or reviewing cancellation and deposit policies for reservations made far in advance. This finding represents an association within the analysed dataset and does not establish that longer lead time directly causes customers to cancel.

9.3 Repeated Guests Show Much Lower Cancellation Rates

The analysis revealed a substantial difference in cancellation behaviour between repeated and non-repeated guests. Repeated guests recorded a cancellation rate of approximately 14.65%, compared with 37.81% for non-repeated guests**.** This indicates that customers who had previously stayed with the hotel were considerably less likely to cancel their reservations within the analysed dataset. From a business perspective, this pattern highlights the potential importance of customer retention and repeat business. Encouraging satisfied guests to return through loyalty initiatives, personalized communication, or targeted offers may help hotels develop a more stable base of customers with historically lower cancellation behaviour. The finding also suggests that repeat-guest status could be a useful factor for hotels to consider when analysing booking reliability and customer behaviour. However, the analysis identifies an association and does not establish that repeat-guest status itself causes lower cancellation rates.

9.4 Customer Type Shows Major Differences in Cancellation Behaviour

Cancellation behaviour varied considerably across different customer types. The analysis produced the following cancellation rates:

- **Transient:** 40.79%
- **Contract:** 30.99%
- **Transient-Party:** 25.45%
- **Group:** 10.10%

Transient customers recorded the highest cancellation rate, while Group customers recorded the lowest. The difference indicates that customer type is an important characteristic when examining

booking reliability and cancellation patterns. From a business perspective, hotels could pay closer attention to Transient bookings when monitoring cancellation risk. Customer-type information could also be considered alongside other factors, such as lead time, hotel type, and booking channel, when developing cancellation-management strategies. At the same time, the comparatively low cancellation rate among Group customers suggests that different customer segments demonstrate substantially different booking behaviours. These differences could be considered when developing segment-specific communication, booking policies, and customer-retention strategies.

9.5 City Hotel Generates Higher Total Completed-Stay Revenue

The revenue analysis identified different financial performance patterns between City Hotel and Resort Hotel. City Hotel generated approximately $14.39 million in completed-stay revenue, compared with approximately $11.60 million for Resort Hotel. Together, the two hotels generated approximately $25.99 million in estimated completed-stay revenue. Although City Hotel generated greater total revenue, Resort Hotel recorded a higher average completed-stay revenue per booking. This indicates that the two hotels have different revenue profiles: City Hotel benefits from a larger overall booking volume, while Resort Hotel generates more completed-stay revenue per booking on average. From a business perspective, total revenue should therefore be considered alongside booking volume and revenue per booking when comparing hotel performance. Focusing only on total revenue could overlook important differences in the value generated by individual completed stays. These results provide useful context for revenue management and demonstrate why multiple performance measures should be considered when evaluating the financial performance of the two hotel types.

9.6 TA/TO Is the Dominant Revenue Distribution Channel

The analysis showed that completed-stay revenue was heavily concentrated within the TA/TO (Travel Agent/Tour Operator) distribution channel. TA/TO accounted for approximately 80.05% of total estimated completed-stay revenue, making it the largest revenue-contributing distribution channel within the analysed dataset. This finding highlights the importance of travel agents and tour operators in generating completed hotel stays and revenue. However, such a high concentration also indicates that a substantial proportion of revenue is associated with a single distribution channel. From a business perspective, hotels could continue to maintain strong relationships with travel agents and tour operators while also monitoring the balance between different booking channels. Developing direct and other distribution channels may provide opportunities to create a more diversified revenue mix. The result does not indicate that reliance on TA/TO is inherently positive or negative, but it identifies distribution-channel concentration as an important area for ongoing revenue analysis.

9.7 Revenue and Pricing Show Seasonal Patterns

The analysis identified seasonal variation in both hotel revenue and pricing throughout the year. Monthly completed-stay revenue changed across different periods, demonstrating that financial performance was not evenly distributed across all months. Seasonal differences were particularly noticeable for Resort Hotel, where Average Daily Rate (ADR) showed stronger variation throughout the year. Pricing increased notably during **July and August**, while City Hotel maintained comparatively more consistent pricing across the analysed period. These patterns indicate that the two hotel types experience different seasonal pricing and revenue dynamics. Resort Hotel appears to have greater seasonal variation, while City Hotel demonstrates a more stable pricing pattern. From a business perspective, understanding these seasonal patterns can support pricing, staffing, capacity planning, and promotional decisions. Hotels could use historical monthly performance to identify periods of stronger and weaker demand and adjust revenue-management strategies accordingly. Seasonal trends

should continue to be monitored over time, as the historical patterns observed in this dataset may not necessarily remain unchanged in future periods.

9.8 Room Changes Are Associated with Different Cancellation Behaviour

The analysis identified a notable difference in cancellation behaviour depending on whether the assigned room type matched the room type originally reserved by the customer. Bookings where the reserved and assigned room types remained the same recorded a higher cancellation rate than bookings where the room type was changed. This pattern was identified by comparing the reserved_room_type and assigned_room_type fields and creating a Room Type Changed indicator during data preparation. From a business perspective, this finding suggests that room-allocation status is worth considering when analysing cancellation behaviour. However, the result should be interpreted carefully because the analysis does not establish that changing a customer's room reduces the likelihood of cancellation. For example, room changes may occur at a different stage of the booking process or may be associated with bookings that were already more likely to result in completed stays. Therefore, additional analysis would be required before drawing conclusions about the effect of room changes on cancellations. Overall, the finding demonstrates how operational booking characteristics can provide additional context when examining cancellation patterns.

9.9 Overall Business Insight

The combined analysis shows that hotel performance is influenced by a combination of booking characteristics, customer behaviour, seasonality, distribution channels, and hotel type rather than by a single factor. Cancellation behaviour emerged as one of the most important areas of the analysis. Higher cancellation rates were associated with City Hotel, longer booking lead times, non-repeated guests, and certain customer types. These patterns provide useful indicators that hotels could monitor when assessing booking reliability and planning future occupancy. The revenue analysis also demonstrated that the two hotel types have different performance profiles. City Hotel generates greater total completed-stay revenue, while Resort Hotel shows stronger seasonal pricing patterns and higher revenue per completed booking. Revenue is also heavily concentrated within the TA/TO distribution channel, demonstrating the importance of booking-source composition when evaluating financial performance. Overall, the findings demonstrate the value of analysing booking, customer, cancellation, and revenue information together. Rather than focusing on a single KPI, hotel management can gain a more complete understanding of performance by considering how these different factors interact across customer segments, booking periods, and hotel types. These insights provide the foundation for developing practical business recommendations aimed at improving cancellation management, customer retention, revenue planning, and distribution strategy.

10. Business Recommendations

Based on the findings from the Python, SQL, and Power BI analysis, several business actions could be considered to improve cancellation management, customer retention, revenue planning, and distribution performance. The recommendations below are based on patterns identified within the historical booking data and should be evaluated alongside operational and commercial considerations before implementation.

10.1 Strengthen Management of Long-Lead-Time Bookings

Booking lead time showed one of the clearest relationships with cancellation behaviour. The cancellation rate increased from approximately 9.60% for bookings made 0–7 days before arrival to

57.01% for bookings made 181 or more days in advance. Hotels could therefore monitor long-lead-time reservations more closely and consider targeted measures such as booking reconfirmation messages, automated reminders, or carefully designed deposit and cancellation policies. Rather than applying the same approach to every reservation, hotels could use lead-time information to identify bookings associated with greater historical cancellation uncertainty and focus communication efforts accordingly. Any changes to deposit or cancellation policies should be tested carefully to ensure that they reduce cancellation uncertainty without unnecessarily discouraging customers from booking in advance.

10.2 Strengthen Repeat-Guest Retention

The analysis showed that repeated guests had a cancellation rate of approximately 14.65%, compared with 37.81% for non-repeated guests. This substantial difference suggests that returning customers represent a comparatively more stable booking segment within the analysed dataset. Hotels could therefore place greater emphasis on encouraging first-time customers to return. Loyalty programmes, personalized follow-up communication, repeat-booking incentives, and targeted offers could be used to strengthen long-term customer relationships. Hotels could also analyse the characteristics of repeated guests in greater detail to understand which booking channels, customer segments, or stay characteristics are most commonly associated with repeat business. These insights could then support more targeted customer-retention strategies. Increasing the proportion of returning customers may help develop a more reliable customer base while also supporting longer-term customer relationships and repeat revenue opportunities.

10.3 Focus Cancellation Management on Higher-Risk Customer Segments

Cancellation rates varied considerably across customer types. Transient customers recorded the highest cancellation rate at approximately 40.79%, compared with 30.99% for Contract customers, 25.45% for Transient-Party customers, and 10.10% for Group customers. Hotels could use customer-type information alongside other booking characteristics, such as lead time and hotel type, to identify reservations associated with greater historical cancellation uncertainty. For higher-cancellation segments such as Transient customers, hotels could consider targeted booking reminders, pre-arrival communication, or reconfirmation processes. Rather than applying additional measures to every customer, this approach would allow cancellation-management efforts to be focused on segments where they may be most relevant. Customer segments should continue to be monitored over time, as cancellation behaviour may change with pricing, booking policies, market conditions, and customer preferences.

10.4 Review City Hotel's Cancellation Strategy

The analysis showed that City Hotel recorded a cancellation rate of approximately 41.79%, compared with 27.77% for Resort Hotel. This indicates that cancellation management is particularly important for City Hotel. Management could investigate City Hotel bookings in greater detail to determine whether the higher cancellation rate is concentrated within particular lead-time categories, customer types, market segments, or distribution channels. This would allow cancellation-management initiatives to be targeted more precisely rather than applying the same approach to all bookings. City Hotel could also monitor cancellation trends regularly through the Power BI dashboard and assess whether changes in booking policies, customer communication, or promotional activity are associated with improvements over time. Because the analysis identifies an association rather than the underlying cause of the higher cancellation rate, further investigation should be conducted before introducing significant policy changes.

10.5 Diversify and Monitor Distribution Channels

The analysis showed that the TA/TO (Travel Agent/Tour Operator) channel accounted for approximately 80.05% of total estimated completed-stay revenue, making it the dominant distribution channel within the dataset. Given this concentration, hotels could continue maintaining strong relationships with travel agents and tour operators while also monitoring the contribution of other booking channels. A highly concentrated revenue mix may make overall performance more dependent on the conditions and performance of a single channel. Hotels could explore opportunities to strengthen direct bookings and other distribution channels through their own websites, loyalty programmes, targeted promotions, or customer-retention initiatives. The objective would not necessarily be to reduce TA/TO bookings, but to develop a more balanced distribution mix while preserving a channel that currently contributes substantial revenue. Revenue contribution, cancellation behaviour, and booking volume should also be monitored together when evaluating distribution channels, as revenue alone may not provide a complete picture of channel performance.

10.6 Use Seasonal Patterns to Support Pricing and Revenue Planning

The analysis identified seasonal variation in monthly revenue and hotel pricing. These patterns were particularly noticeable for Resort Hotel, where ADR showed stronger seasonal changes, especially during July and August, while City Hotel maintained comparatively more consistent pricing. Hotels could use historical seasonal patterns to support pricing, staffing, promotional, and capacity-planning decisions. Periods associated with stronger demand and higher ADR may provide opportunities to optimize pricing, while lower-performing periods could be supported through targeted promotions or booking incentives. City Hotel and Resort Hotel should also be considered separately when developing seasonal strategies because their pricing and revenue patterns differ. Applying the same pricing approach to both properties may overlook these differences. Regular monitoring of monthly bookings, ADR, cancellations, and completed-stay revenue through the Power BI dashboard could help management identify changes in seasonal performance and adjust strategies accordingly.

10.7 Use the Dashboard for Ongoing Performance Monitoring

The Power BI dashboard developed in this project provides a structured way to monitor booking, cancellation, and revenue performance across different hotels and time periods. Hotel management could use similar dashboard reporting to regularly track key performance indicators such as total bookings, cancellation rate, completed-stay revenue, and Average Daily Rate (ADR). The Hotel, Year, and Month filters also allow performance to be examined at a more detailed level when required. Particular attention could be given to indicators identified as important during the analysis, including long-lead-time bookings, customer type, repeat-guest status, seasonal changes, and distribution-channel performance. Regular monitoring would allow management to identify changes in booking and revenue patterns earlier and evaluate whether operational, pricing, customer-retention, or cancellation-management initiatives are producing the intended results. Overall, integrating interactive dashboard reporting into ongoing performance reviews could support faster and more consistent data-driven decision-making.

11. Limitations

Although the project provides useful insights into hotel booking, cancellation, customer, and revenue patterns, several limitations should be considered when interpreting the results.

11.1 Historical Dataset

The analysis is based on historical booking data covering arrivals from July 2015 to August 2017. Customer behaviour, hotel pricing, distribution channels, booking platforms, and market conditions may have changed since this period. Therefore, the results should be interpreted as patterns within the analysed dataset rather than as a representation of current hotel-market conditions.

11.2 Limited Hotel Coverage

The dataset contains booking information for only two hotel types: City Hotel and Resort Hotel. The findings therefore cannot automatically be generalized to all hotels, locations, or hospitality businesses.

11.3 Estimated Revenue

Revenue in this project was calculated using ADR × Total Nights and was treated as completed-stay revenue when the reservation status indicated a completed stay. Therefore, the revenue figures should be interpreted as analytical estimates rather than audited financial revenue. The calculation does not represent every possible source of hotel revenue or adjustment, such as food and beverage spending, additional services, discounts, commissions, taxes, refunds, or other charges.

11.4 Association Does Not Establish Causation

Several relationships were identified between cancellation behaviour and characteristics such as booking lead time, hotel type, customer type, repeat-guest status, and room-change status. These findings demonstrate associations within the dataset but do not establish that these characteristics directly caused customers to cancel their bookings. Additional analysis and operational context would be required to determine causal relationships.

11.5 Data Quality and Assumptions

The dataset contained unusual and missing values that required analytical decisions during the Python data-preparation stage. For example, an extreme ADR value was treated as missing because it could not be independently verified. Although these decisions were documented and designed to avoid introducing unsupported assumptions, data-cleaning choices can still influence analytical results. Overall, these limitations should be considered when using the findings for business decision-making or applying them to other hotel environments.

12. Conclusion

The Hotel Booking Analytics project demonstrated an end-to-end data analytics workflow for transforming raw hotel booking data into meaningful business insights using Python, SQL, and Power BI.

Python was used to inspect, clean, transform, and explore the dataset, creating a reliable foundation for analysis. SQL was then used to validate the prepared data and investigate business-focused questions relating to bookings, cancellations, customer behaviour, seasonality, and revenue. Finally, Power BI was used to transform the analytical findings into an interactive three-page dashboard covering executive performance, booking and cancellation behaviour, and revenue and seasonal trends.

The analysis identified several important patterns. Cancellation rates were higher for City Hotel, long-lead-time bookings, non-repeated guests, and certain customer types. Revenue analysis showed that City Hotel generated higher total completed-stay revenue, while Resort Hotel demonstrated a different pricing and revenue profile. The analysis also highlighted seasonal variation and a strong concentration of completed-stay revenue within the TA/TO distribution channel.

These findings demonstrate how hotel booking data can support more informed cancellation management, customer-retention initiatives, revenue planning, and distribution-channel analysis.

Overall, the project demonstrates the practical integration of data preparation, exploratory analysis, SQL querying, business analysis, and interactive visualization to convert raw data into clear and actionable business insights.

13. Tools & Technologies

The project used three core data analytics technologies—Python, SQL, and Power BI—with each tool supporting a different stage of the analytical workflow.

13.1 Python

Python was used for data inspection, cleaning, transformation, exploratory data analysis, and preparation of the final analytical dataset.

The main Python libraries used were:

- **Pandas** – data manipulation, cleaning, aggregation, and analysis
- **NumPy** – numerical operations and data handling
- **Matplotlib** – data visualization
- **Seaborn** – statistical and exploratory visualizations

The Python analysis was developed in **Google Colab**, providing a notebook-based environment for combining code, outputs, visualizations, and analytical observations.

13.2 SQL

SQL was used to perform structured analysis on the cleaned dataset and answer business-focused questions relating to:

- Booking and cancellation performance
- Customer behaviour
- Lead-time patterns
- Seasonal trends
- Revenue performance
- Market segments
- Distribution channels

SQL techniques used throughout the analysis included filtering, aggregation, conditional calculations, grouping, ordering, and ranking to transform booking records into business-level insights.

13.3 Power BI

Power BI was used as the final data visualization and business intelligence layer of the project.

The final dashboard consisted of three pages:

- **Executive Overview**
- **Booking & Cancellation Analysis**
- **Revenue & Seasonal Trends**

The dashboard incorporated KPI cards, comparative charts, trend visualizations, and interactive Hotel, Year, and Month slicers to allow users to explore the analytical results dynamically.

13.4 End-to-End Workflow

The overall project workflow followed the sequence:

Raw Hotel Booking Data → Python Data Preparation & EDA → SQL Business Analysis → Power BI Dashboard → Business Insights & Recommendations

This workflow demonstrates the integration of multiple analytics tools to progress from raw data preparation through detailed analysis and finally to interactive business reporting.

14. References

Original Dataset Publication

António, N., de Almeida, A., & Nunes, L. (2019). *Hotel booking demand datasets*. **Data in Brief, 22**, 41–49. https://doi.org/10.1016/j.dib.2018.11.126

Kaggle Data Source

Arunkumar V. EDA with Hotel Booking Demand ML & DL. Kaggle Notebook. https://www.kaggle.com/code/arunkumarpyramid/eda-with-hotel-booking-demand-ml-dl/notebook

The Hotel Booking Demand dataset used in this project was accessed through the Kaggle source above. The underlying dataset originates from the Hotel Booking Demand datasets published by António, de Almeida, and Nunes (2019).